ANSWER KEY & MARKING SCHEME SET A

Chapters 9 & 10: **The Baudhayana–Pythagoras Theorem, Proportional Reasoning-2** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 17 cm	$\sqrt{8^2+15^2}=17$
2	**(a)** 10 cm	$\sqrt{6^2+8^2}=10$
3	**(d)** 15 cm	$\sqrt{17^2-8^2}=15$
4	**(c)** Yes	$6^2+8^2 = 10^2$
5	**(b)** Yes	$14^2+48^2 = 50^2$
6	**(c)** 450 km	Multiply by the scale
7	**(a)** ₹ 200	One part = ₹ 100
8	**(b)** ₹ 240	One part = ₹ 60
9	**(a)** 6	Workers × days is constant
10	**(c)** 90°	Angle = percent × 3.6

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Other leg 15 cm (1); area 60 cm² (1).	1+1
12	PR (2).	2
13	4 hours (2).	2
14	126° (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Diagonal 15 m (1.5); saved $21-15=6$ m (1.5).	1½+1½
16	Total work $=300$ worker-days (1); days $=12$ (1); difference 8 days (1).	1+1+1
17	Diagonal 29 m (1.5); saved $41-29=12$ m (1.5).	1½+1½
	OR: Triplet $(3,4,5)$ (1.5); $9+16=25=25$ (1.5).	1½+1½
18	Total work $=432$ worker-days (1); days $=12$ (1); difference 6 days (1).	1+1+1
	OR: Total work $=300$ worker-days (1); days $=25$ (1); difference 5 days (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) First foot distance 8 m; second 15 m (3). (ii) Moved 7 m away from the wall (2).	3+2
	OR: (i) 100 m (2). (ii) $\sqrt{150^2-100^2}\approx 111.8$ m (2). (iii) (75, 100, 125) i.e. (3, 4, 5)×25 (1).	2+2+1
20	(i) 20 m/s (2). (ii) $750/20=37.5$ s (3).	2+3
	OR: (i) 9 L (2). (ii) 9 : 25 (3).	2+3

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapters 9 & 10: **The Baudhayana–Pythagoras Theorem, Proportional Reasoning-2** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** $10\sqrt{2}$ cm	Hypotenuse of isosceles right triangle
2	**(b)** $a\sqrt{2}$	Diagonal of the original square
3	**(c)** obtuse-angled	Converse idea
4	**(d)** 6 m	$\sqrt{100-64}$
5	**(b)** 13	$\sqrt{25+144}$
6	**(c)** 108°	Angle = percent × 3.6
7	**(c)** 360°	Full turn
8	**(c)** 4 hours	Pipes × hours = 60
9	**(c)** 25%	90/360
10	**(a)** inverse proportion	Definition

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	(6, 8, 10), (8, 15, 17); (5, 12, 13) (2).	2
12	The diagonal of a rectangle produces an area equal to the sum of the areas produced by its length and breadth (2).	2
13	15 cm (2).	2
14	$x = 12$, $y = 15$ (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Triplet $(5, 12, 13)$ (1.5); $25 + 144 = 169 = 169$ (1.5).	1½+1½
16	Total work $= 360$ worker-days (1); days $= 20$ (1); difference 25 days (1).	1+1+1
17	Triplet $(7, 24, 25)$ (1.5); $49 + 576 = 625 = 625$ (1.5).	1½+1½
	OR: Triplet $(9, 40, 41)$ (1.5); $81 + 1600 = 1681 = 1681$ (1.5).	1½+1½
18	Tea 100, coffee 80, juice 60 (3).	3
	OR: Ratio 2 : 3 (1); ₹ 3000 and ₹ 4500 (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Height $\sqrt{13^2 - 5^2} = 12$ cm (3). (ii) Area $\frac{1}{2}(14 + 24) \times 12 = 228$ cm² (2).	3+2
	OR: (i) The diagonal's square equals the sum of squares of the sides (2). (ii) 37 (3).	2+3
20	(i) 750 g (3). (ii) Direct (2).	3+2
	OR: (i) 6000, 4000, 10000 (2). (ii) 108°, 72°, 180° (2). (iii) 50% (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET C

Chapters 9 & 10: **The Baudhayana–Pythagoras Theorem, Proportional Reasoning-2** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 25 cm	$\sqrt{7^2+24^2}=25$
2	**(b)** 20 cm	$\sqrt{12^2+16^2}=20$
3	**(d)** 24 cm	$\sqrt{25^2-7^2}=24$
4	**(d)** No	$9^2+12^2 \neq 16^2$
5	**(c)** No	$10^2+11^2 \neq 15^2$
6	**(b)** 100 km	Multiply by the scale
7	**(d)** ₹ 300	One part = ₹ 100
8	**(a)** ₹ 240	One part = ₹ 80
9	**(c)** 21	Workers × days is constant
10	**(b)** 36°	Angle = percent × 3.6

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$25+36=61 \neq 64$, no (2).	2
12	29 m (2).	2
13	8 days (2).	2
14	Direct: ratio constant; inverse: product constant (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\sqrt{225-81}=12$ m (3).	3
16	1 cm = 4 km (1); $1:400000$ (2).	1+2
17	Side 17 cm (2); perimeter 68 cm (1).	2+1
	OR: $225+400=625$ (2); area 150 cm² (1).	2+1
18	20 days (3).	3
	OR: 90°, 126°, 144° (1.5); 90, 126, 144 students (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Half diagonals form a right triangle, so $(d_1^2+d_2^2)/4=s^2$ (3). (ii) $36+64=100=4\times25$ (2).	3+2
	OR: (i) 50 m (2). (ii) ₹ 2000 (1). (iii) 62 m, saving 12 m (2).	2+1+2
20	(i) 5 hours (2). (ii) 7.5 hours (3).	2+3
	OR: (i) 90 man-days (2). (ii) Half is 45, so 15 men (3).	2+3

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET D

Chapters 9 & 10: **The Baudhayana–Pythagoras Theorem, Proportional Reasoning-2** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** $12\sqrt{2}$ cm	Hypotenuse of isosceles right triangle
2	**(d)** three numbers satisfying $a^2+b^2=c^2$	Definition
3	**(b)** (3, 4, 5)	m=2, n=1
4	**(d)** 17 cm	$\sqrt{64+225}$
5	**(b)** $5\sqrt{2}$ cm	$d/\sqrt{2}$
6	**(b)** 1 km	100000 cm = 1 km
7	**(b)** 2 : 3 : 5	B is already common
8	**(c)** 4 cm	200 cm / 50
9	**(a)** 2 : 3 : 5	In proportion to investment
10	**(b)** 1/2	3 of 6 parts

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$5\sqrt{2}$ cm (2).	2
12	$7\sqrt{2}\approx 9.9$ cm (2).	2
13	75 (2).	2
14	₹ 240 (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	13 m (3).	3
16	A is faster (1); daily work ratio $\frac{1}{12}:\frac{1}{15}=5:4$ (2).	1+2
17	Other side 9 cm (1); diagonal $\sqrt{145}\approx 12.04$ cm (2).	1+2
	OR: $4+9\neq 16$ (1.5); $81+144=225$ (1.5).	1½+1½
18	Sum 180° (1); 40°, 60°, 80° (2).	1+2
	OR: 5 parts = 35, one part 7 (1.5); 49 and 14 (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $r=\sqrt{12^2+5^2}=13$ cm (3). (ii) Perpendicular from centre bisects the chord (2).	3+2
	OR: (i) Statement and use in checking corners of a rectangle (3). (ii) $256+3969=4225=65^2$ (2).	3+2
20	(i) Make B = 20 (2). (ii) 30, 40, 48 (3).	2+3
	OR: (i) 10 m × 7 m (2). (ii) 70 m² (1). (iii) ₹ 31500 (2).	2+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.